

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is the most consequential validity gap. The benchmark's NER label spaces (UMLS Semantic Groups in QUAERO and MANTRAGSC [Q35, Q103, Q106]; CLINENTITY in E3C [Q104]; SUBSTANCE/DOSAGE/MODE/FREQUENCY in DEFT2021 [Q108]; ICD-10 chapters in DiaMed [Q107]; MeSH Chapter C in DEFT2021 cls [Q27]) are clinical/research ontologies that do not map to the regulatory entity types (INNs distinguished from brand names, ATC codes, excipient nomenclature, MedDRA preferred terms, EMA posology template fields) required by the deployment. Dataset analysis confirms the CHEM label conflates INN, brand name, excipient, and endogenous chemical (QUAERO-D13, QUAERO-D14, MANTRAGSC-D11); drug names appear untagged in E3C (E3C-D12) and CAS (CAS-D15); DEFT2021 SUBSTANCE conflates INN/excipient/chemical (DEFT2021-D10). For STS, the 0–5 Likert scale [Q22, Q41] is not calibrated for regulatory equivalence: dataset analysis demonstrates that the rubric assigns score 4.0 to pairs differing in precautionary qualifiers (DEFT2020-D2, DEFT2020-D18), administration route additions (DEFT2020-D5), twofold quantitative differences (CLISTER-D17, CLISTER-D18), and brand-vs-general formulations (DEFT2020-D19) — exactly the small-magnitude differences that trigger Type IA/IB/II variation submissions under EMA/ANSM rules [WEB-6]. Web search confirmed no published crosswalk between DrBenchmark NER schemas and EMA/ANSM regulatory annotation guidelines [WEB-12].

ID	Evidence
Q35	"10 entity categories corresponding to the UMLS Semantic Groups (Lindberg et al., 1993) were annotated (more detail in Appendix B.3)." (p.3)
Q103	"O, GEOG, PHEN, DISO, ANAT, OBJC, PHYS, PROC, DEVI, CHEM and LIVB" (p.14)
Q106	"Medline: ANAT, PROC, CHEM, PHYS, GEOG, DEVI, LIVB, OBJC, DISO, PHEN and O. EMEA and Patents: ANAT, PROC, CHEM, PHYS, DEVI, LIVB, OBJC, DISO, PHEN and O." (p.14)
Q104	"Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML" (p.14)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
Q107	"Multi-label Classification: immunitaire (immunology), endocriniennes (endocrinology), blessures (injury), chimiques (chemicals), etatsosy (signs and symptoms), nutritionnelles (nutrition), infections (infections), virales (virology), parasitaires (parasitology), tumeur (oncology), osteomusculaires (osteomuscular disorders), stomatognathique (stomatology), digestif (digestive system disorders), respiratoire (respiratory system disorders), ORL (otorhinolaryngologic diseases), nerveux (nervous system disorders), oeil (eye diseases), homme (male genital diseases), femme (female genital diseases), cardiovasculaires (cardiology), hemopathies (hemic and lymphatic diseases), genetique (genertic disorders) and peau (dermatology)." (p.14)
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q22	"The first task aims at identifying the degree of similarity within pairs of sentences, from 0 (the less similar) to 5 (the most similar)." (p.3)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
WEB-6	Variation types IA/IB/II directly map to threshold-based escalation requirements but are not evaluated in benchmark (https://alhena-consult.com/how-to-market-a-drug-in-france/)

WEB-12	No French regulatory NER benchmark exists; English ADE Eval required modification before regulatory use (https://pubmed.ncbi.nlm.nih.gov/29860093/)
QUAERO-D13	Refludan 50 mg en poudre pour solution injectable ... Lepirudine Les autres composants sont le mannitol (E 421) et l' hydroxyde de sodium — Brand name, INN, excipients all share CHEM label — regulatory distinction collapsed
QUAERO-D14	Phosphate de sodium , monobasique ... Polysorbate 80 (E433) ... Eau pour préparation injectable — Excipients tagged identically to active ingredients under CHEM
MANTRAGSC-D11	“antirétroviraux” (tag=2, CHEM) — Drug class labeled same as INN; no regulatory sub-distinction
E3C-D12	mis sous traitement par ciclosporine — Drug name tagged O; schema lacks INN/substance entity type required for regulatory NER
CAS-D15	"administration d'une ampoule de digoxine en intraveineuse" ner_tags: [0,...,0] all zeros — Drug name entirely untagged; NER only captures negation scope, not pharmacological entities
DEFT2021-D10	diphenhydramine et de méthylprednisolone — Drug names tagged SUBSTANCE with no INN/ATC/excipient distinction
DEFT2020-D2	En conséquence, par mesure de précaution, il convient d'éviter d'allaiter pendant la durée du traitement. — Breastfeeding contraindication with precautionary qualifier; cible drops qualifier
DEFT2020-D18	scores=[5.0, 2.0, 4.0, 4.0, 5.0] — Precautionary qualifier dropped in cible; rated 4.0 — legally significant difference not captured
DEFT2020-D5	Comprimé rond, blanc, biconvexe, gravé 6 sur une face, une flèche étant gravée sur l'autre face. — Pharmaceutical form description; cible adds "sublingual" route — regulatory-significant difference
CLISTER-D17	Le recul est de 2 ans." / "Le suivie est de 4 ans et demi. — Different follow-up durations scored 4.0; same rubric would conflate regulatory dose differences
CLISTER-D18	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — Twofold quantitative difference scored 4.0; regulatory dose doubling requires near-0
DEFT2020-D19	ce médicament est contre-indiqué dans les cas suivants" vs. "n'utilisez jamais diclofenac eg 1%, gel dans les cas suivants — General vs. brand-specific prohibition; general STS would conflate

Strengths

- DEFT2021 NER provides 13 fine-grained entity types including SUBSTANCE, DOSAGE, MODE, FREQUENCY [Q28, Q108] — partial overlap with posology entity types (DEFT2021-D5)
- PxCorpus NER covers 38 prescription-related entity classes including drug, dose, mode (PXCORPUS-D1 through PXCORPUS-D7) [Q44]
- STS uses Spearman + EDRM dual metrics [Q54], providing some sensitivity analysis capability
- DEFT2020 retains individual annotator scores enabling uncertainty quantification (DEFT2020-D12, DEFT2020-D13)

ID	Evidence
Q28	“The second task involves fine-grained information extraction for 13 types of entities (more detail in Appendix B.7).” (p.3)
Q108	“Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE” (p.14)
DEFT2021-D5	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Posology-adjacent annotation of drug, dose, route, frequency
Q44	“PxCorpus (Kocabiyikoglu et al., 2022) is a spoken language understanding dataset in the domain of medical drug prescription transcripts. It includes 4 hours (1,981 recordings) of transcribed and annotated dialogues focused on drug prescriptions. The recordings were manually transcribed and semantically annotated. The first task involves classifying the textual utterances into one of the 4 intent classes (prescribe, replace, negate, none). The second task is a NER task where each word in a sequence is classified into one of 38 classes, such as drug, dose, or mode (more detail in Appendix B.9).” (p.4)
PXCORPUS-D 1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Brand drug with dose, form, indication, frequency, duration — deployment-relevant posology structure
PXCORPUS-D 7	discotrine 10 milligrammes patch 1 patch le matin au réveil retirer le patch le soir pendant 1 mois — Transdermal patch with timed application/removal
Q54	“For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020).” (p.6)
DEFT2020-D12	scores=[5.0, 2.0, 0.0, 3.0, 1.0] — High inter-annotator disagreement; signals annotation uncertainty
DEFT2020-D13	scores=[2.0, 4.5, 0.0, 1.0, 0.0] — Extreme annotator divergence on low-similarity pair

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are predominantly misaligned with regional regulatory relevance. UMLS Semantic Groups [Q35], ICD-10 chapters [Q107], MeSH C axes [Q27], and CLINENTITY [Q104] reflect clinical/research ontologies, not EMA/ANSM regulatory labeling ontologies.

OO-2: Yes — multiple regionally-required categories are missing: INN-vs-brand distinction, ATC codes, excipient nomenclature as distinct from active ingredients, MedDRA preferred terms, EMA posology template fields (population/dose/route/frequency as separately typed), contraindication qualifier types, regulatory variation type (IA/IB/II) classifications. Confirmed by dataset analysis (QUAERO-D13, MANTRAGSC-D11, DEFT2021-D10) and web search [WEB-12].

OO-3: The CHEM label collapses regulatory-relevant distinctions (INN, brand, excipient, endogenous chemical) under a single category — encoding a clinical-research assumption that pharmaceutical sub-types are not separately operationally relevant (QUAERO-D13, QUAERO-D14, MANTRAGSC-D11). The 0–5 STS scale [Q22] encodes general semantic proximity rather than regulatory equivalence, treating legally significant qualifier differences as high-similarity (DEFT2020-D2, DEFT2020-D18, CLISTER-D17).

OO-4: Stakeholder-driven taxonomy redesign is necessary: regulatory affairs experts would need to define INN/brand/excipient/MedDRA-aligned NER schema and a regulatory-equivalence STS rubric calibrated to variation types (IA/IB/II) [WEB-6]. No such schema exists in DrBenchmark.

OO-5: Severe structural and content validity violations: NER schemas omit regulatory entity sub-types; STS scoring treats legally significant variants as equivalent. The benchmark cannot validly evaluate models for regulatory compliance decisions without taxonomy redesign.

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WEB-11	MedDRA available in French translation but no French MedDRA NER benchmark (https://bioportal.bioontology.org/ontologies/MEDDRA)

Information Gaps

- No empirical demonstration of how a model fine-tuned on benchmark NER labels would perform when re-evaluated against a regulatory-aligned schema

Requires Expert Verification

- Whether DEFT2021's 13-type NER scheme (SUBSTANCE/DOSAGE/MODE/FREQUENCY) could serve as a partial regulatory-NER training signal after schema mapping by regulatory affairs experts
- Specific STS score thresholds that map to Type IA/IB/II variation classifications

Remediation

Gap 1: NER label spaces collapse INN, brand name, excipient, and endogenous chemical under single labels (CHEM, SUBSTANCE), and the 0–5 STS scale is not calibrated for regulatory equivalence under EMA/ANSM Type IA/IB/II variation rules.

Recommendation: Construct a regulatory-aligned NER schema (INN, brand, excipient, ATC, MedDRA PT, posology fields) and a regulatory-equivalence STS rubric mapped to variation types, validated by regulatory affairs specialists. Re-evaluate top-performing benchmark models against this supplementary schema before deployment use.